PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

Lockbook

Group Members

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Project Guide
Mr.Kiran Gaikwad

Contents

- Introduction
- Objectives
- Scope
- Features / Functionality
- Project Outcomes
- Technology Stack
- Block Diagram if applicable

1. Introduction

LockBook is a secure password management application designed to store and manage multiple account credentials safely.
It provides centralized access to passwords using master authentication.
The system improves both security and user convenience.

Problem Identified

1.Users struggle to remember multiple passwords.
2.Many people reuse weak passwords, increasing security risk.
3.Manual password entry is time-consuming and inconvenient.

Solution Proposed

1.A secure master login system for controlled access.
2.Encrypted password vault to store credentials safely.
3.Get access for faster and secure login

2. Objectives

List of objectives:

1. User Authentication & Master Access:

To design and implement a secure master authentication system that controls access to the password vault.

2. Password Vault & Storage Module

To create a secure password vault that stores multiple account credentials in encrypted form.

3. Secure Open-Link Access

To enable controlled access to saved credentials through secure open links, allowing users access that app

4. Security, Backup & Activity Module

To enhance overall system security and maintain password data reliability.

3. Scope

- Can be used for personal use to securely store and manage passwords.
- Can be used by students and teachers to manage academic accounts safely.
- Can be used in organisations and offices to protect employee credentials.
- Can be used by professionals and freelancers managing multiple online accounts.
- Can be used in mobile and web applications with cloud access and advanced security features.

4. Feature /Functionality

Feature 1 : Master Login Authentication

Provides a single secure master login that allows users to access all their stored passwords. It ensures only authorized users can enter the system using strong authentication methods.

Feature 2 : Get Access

Enables users to securely access their stored passwords and account information after login.

Feature 3 : Password Management (CRUD Operations)

Allows users to view, add, update, and delete saved passwords easily. This helps in organising and managing multiple account credentials in one place.

4. Feature /Functionality

Feature 4 : Secure Encrypted Storage

Stores all passwords in a secure local database using encryption techniques. This ensures that sensitive data remains protected from unauthorised access.

Feature 5 : Secondary Device Access Notification

Sends alerts to the user when their passwords are accessed from another device. This feature increases security by informing users about any unusual activity.

Feature 6 : User-Friendly Interface

Provides a simple and easy-to-use interface for better user experience. It helps users navigate and manage their passwords without confusion.

5. Outcome of Project

- User can create and secure the application using a Master Password.
- User can safely store multiple account usernames and passwords.
- User can view saved credentials only after successful master authentication.
- User can update or delete stored passwords anytime.
- All passwords are stored in encrypted format for better security.
- User can quickly search and manage saved account details in one place.

6. Technology Stack

Front-End (GUI): Flutter

- Design a simple and attractive user interface for LockBook with screens like login, dashboard, add/edit password, and settings.

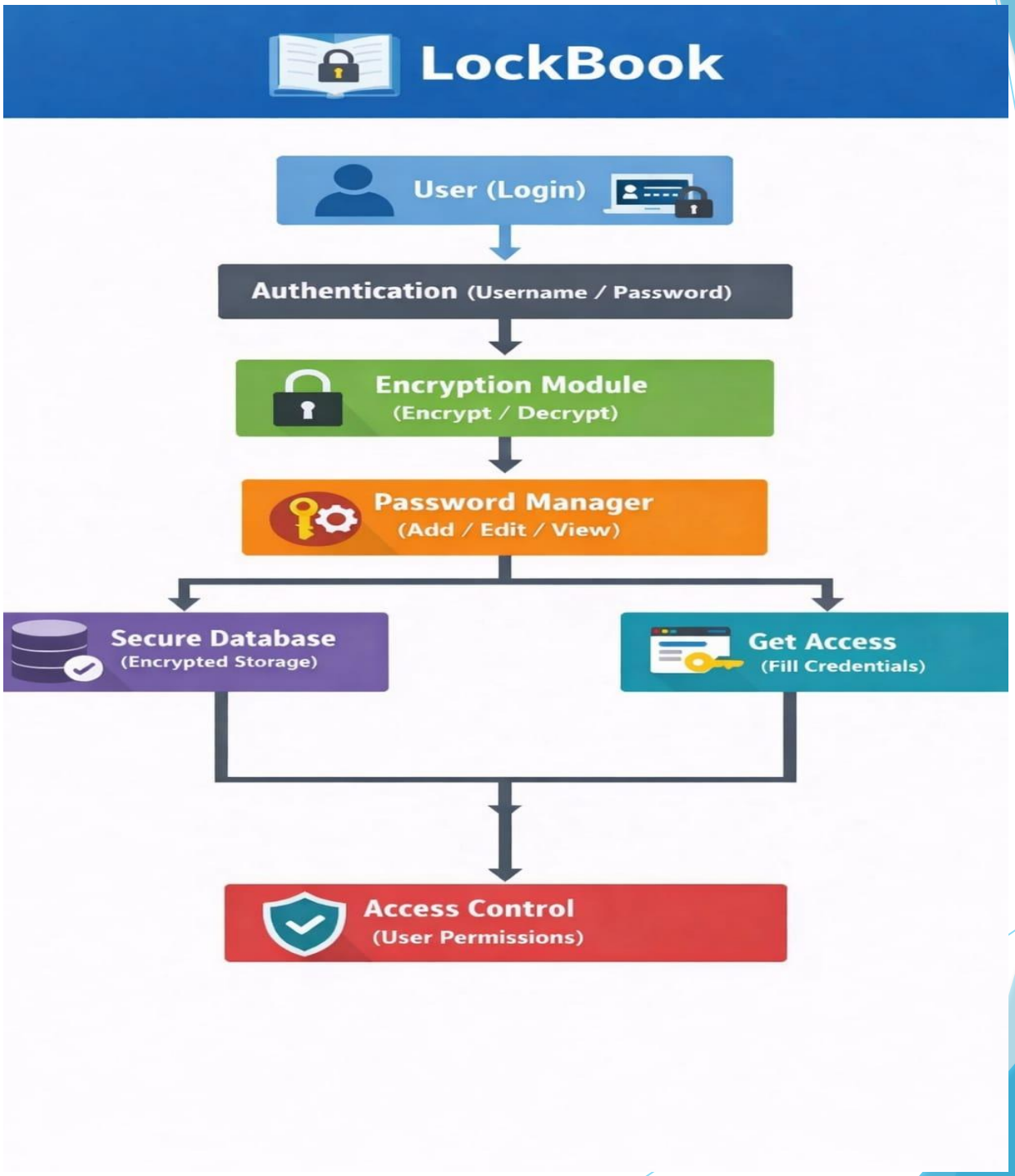
- Implement features like search, auto-fill option, and easy navigation so users can quickly manage their stored passwords.

Back-End (Database): Firebase

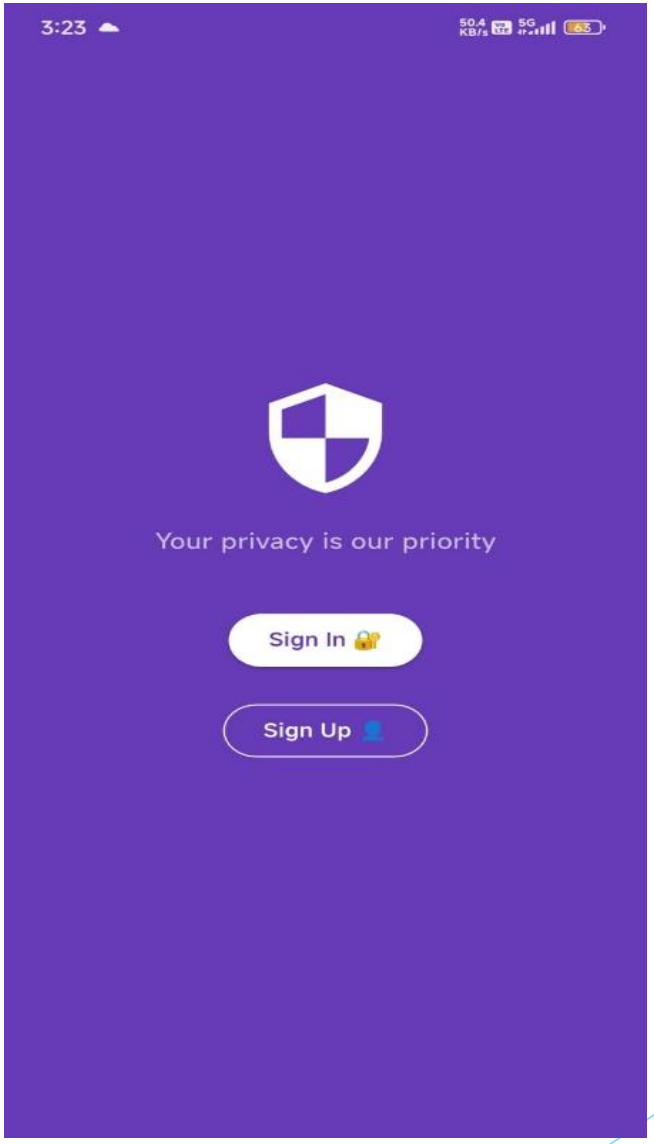
- Develop a secure backend with an encrypted local database to store user credentials and password data safely.

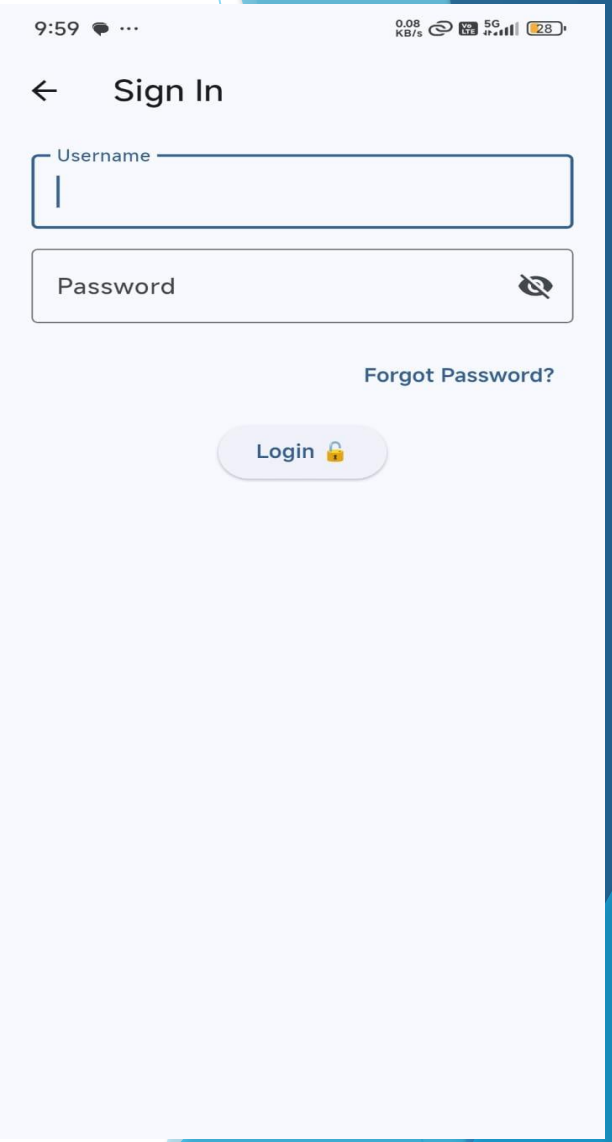
- Implement authentication, encryption/decryption logic, and access control to ensure data security and proper functioning of the system.

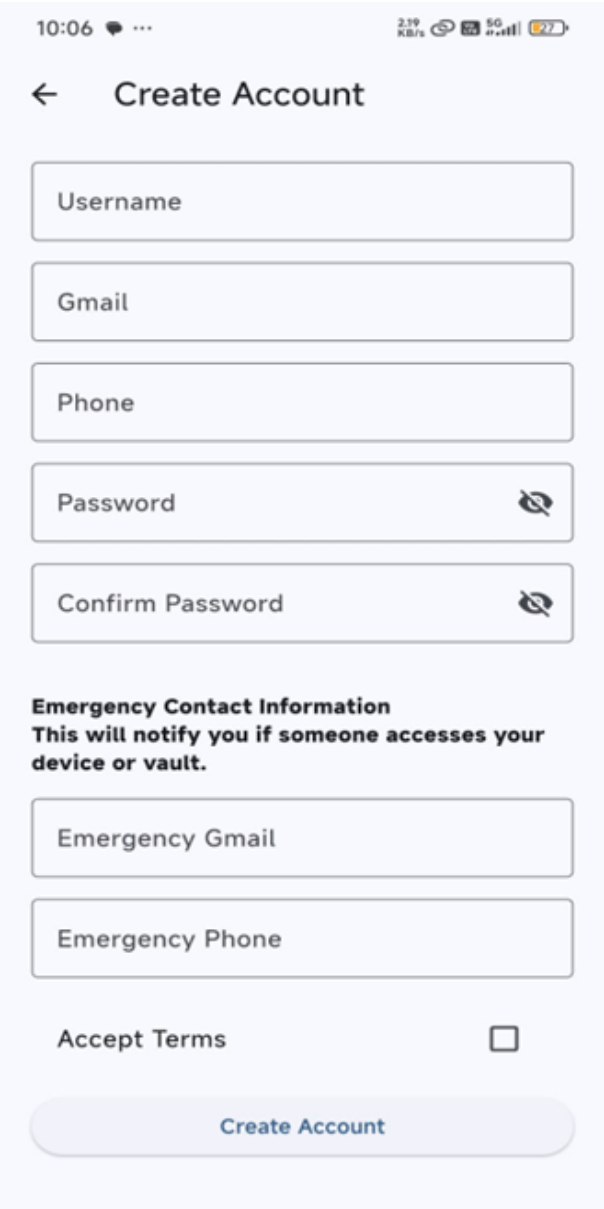
7. Block Diagram (if applicable)

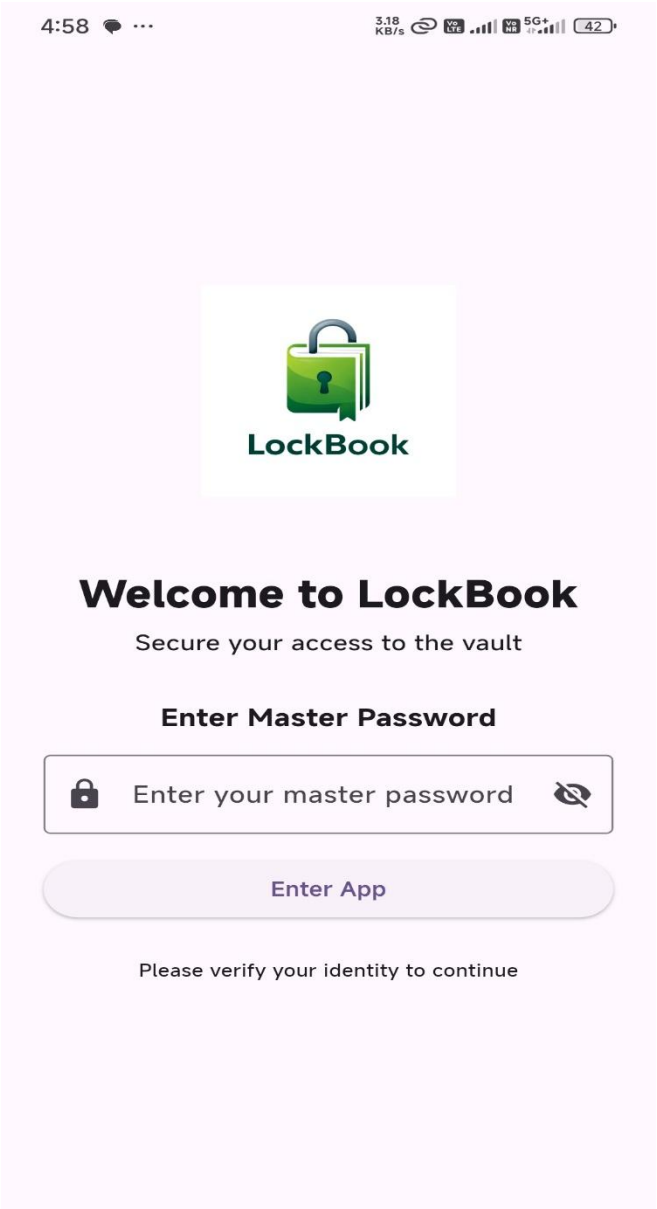


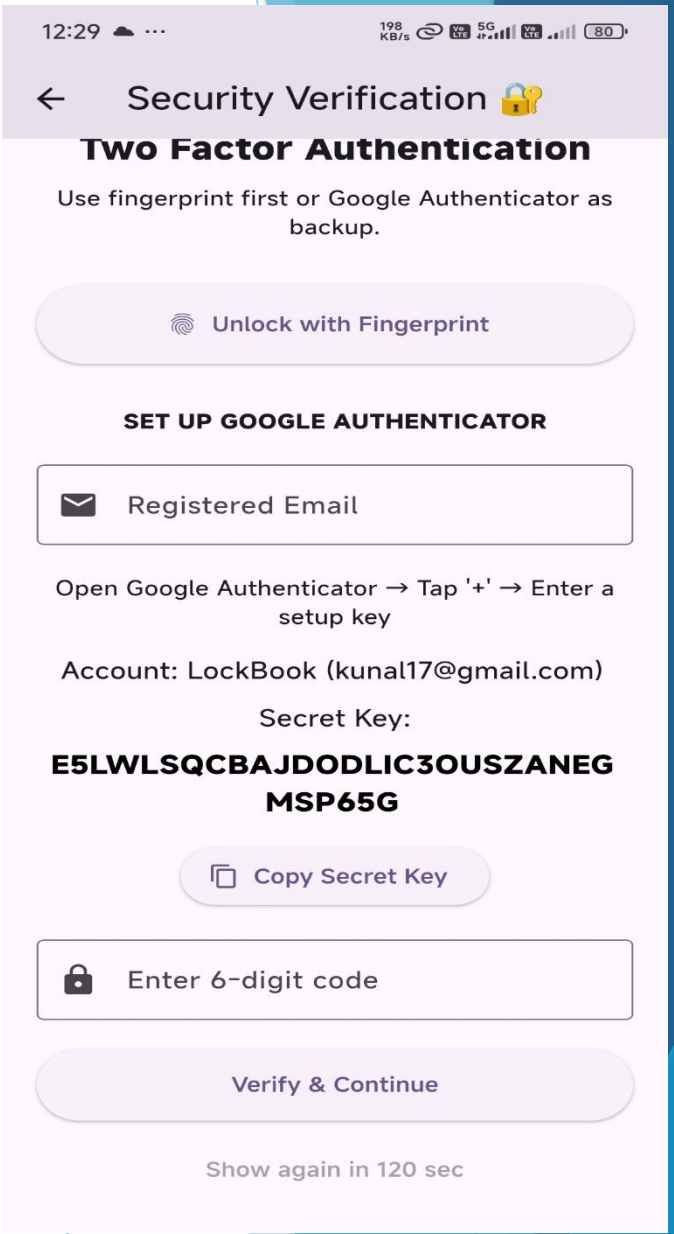
8. Frontend & Backend

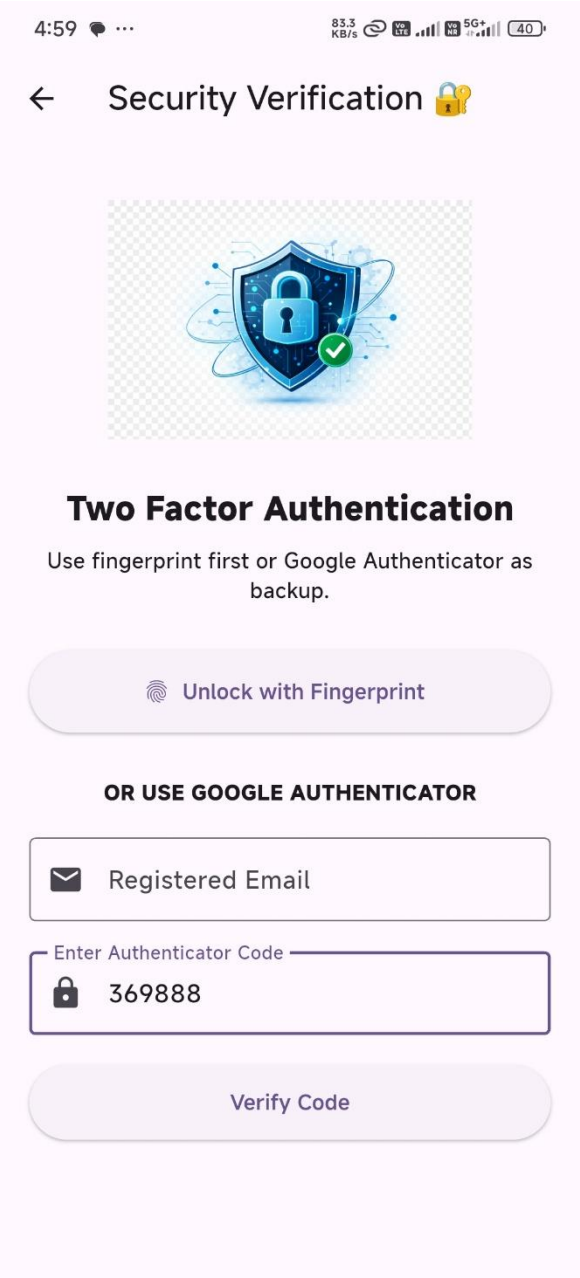


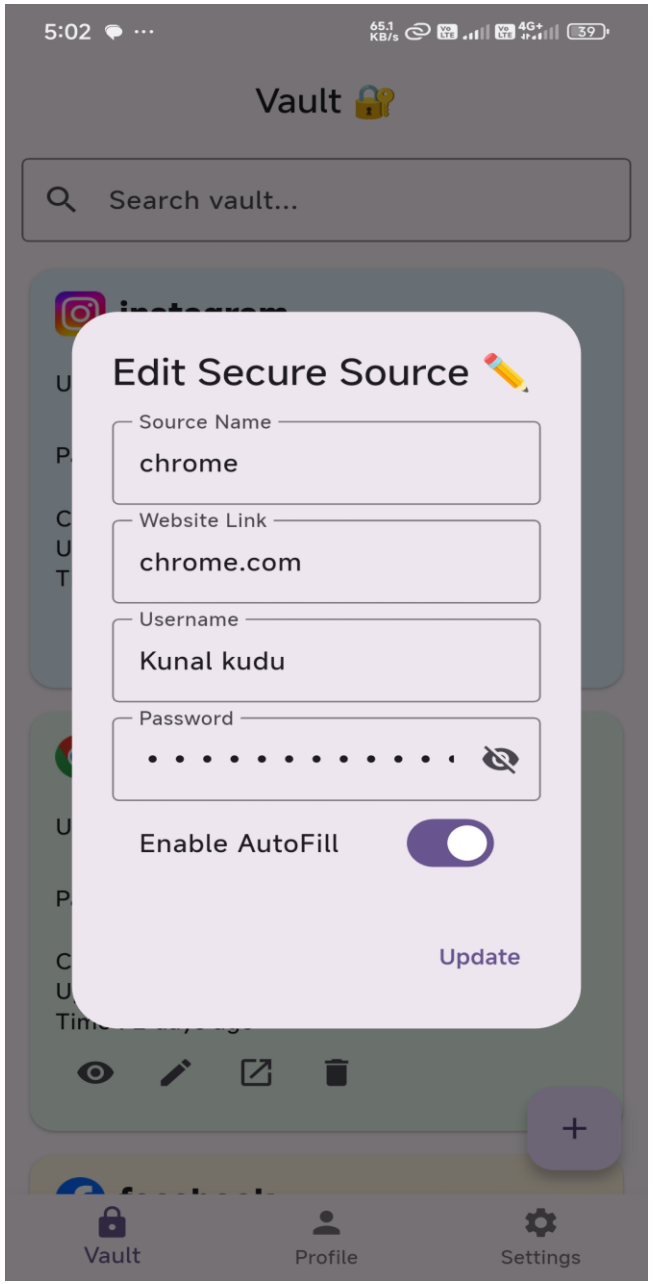


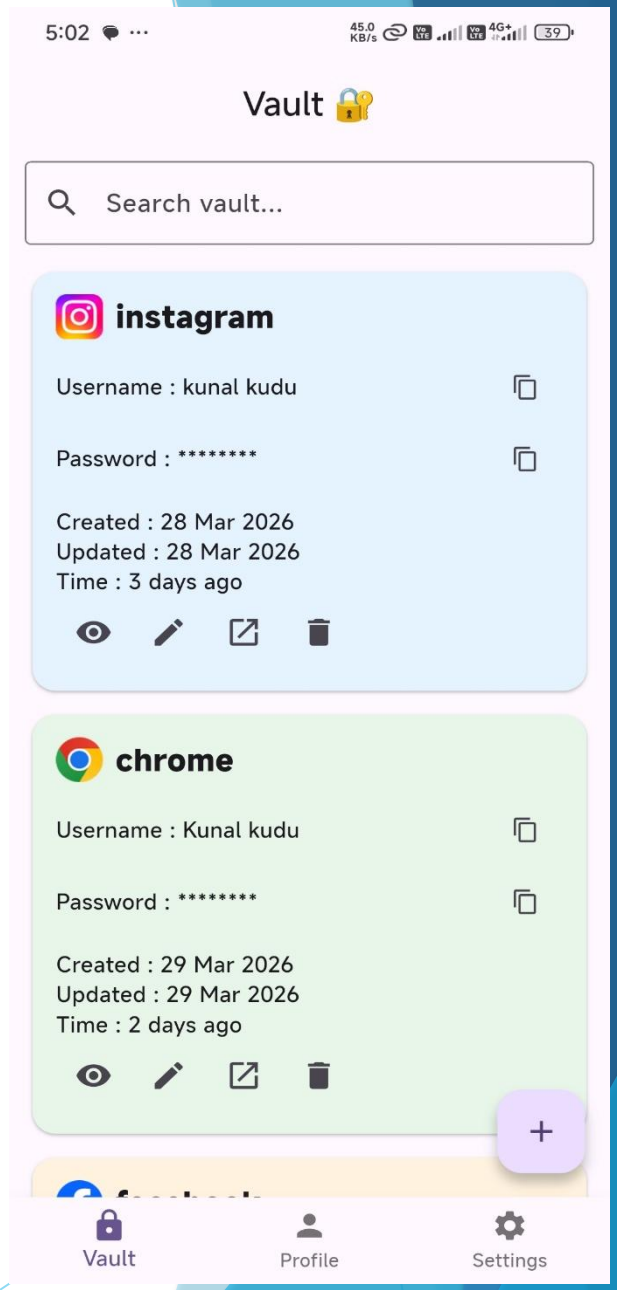

10:06
Create Account
Username
Gmail
Phone
Password
Confirm Password
Emergency Contact Information
This will notify you if someone accesses your device or vault.
Emergency Gmail
Emergency Phone
Accept Terms
Create Account


4:58
LockBook
Welcome to LockBook
Secure your access to the vault
Enter Master Password
Enter your master password
Enter App
Please verify your identity to continue


12:29
Security Verification
Two Factor Authentication
Use fingerprint first or Google Authenticator as backup.
Unlock with Fingerprint
SET UP GOOGLE AUTHENTICATOR
Registered Email
Open Google Authenticator → Tap '+' → Enter a setup key
Account: LockBook (kunal17@gmail.com)
Secret Key:
E5LWLSQCBAJDODLIC3OUSZANEGMSP65G
Copy Secret Key
Enter 6-digit code
Verify & Continue
Show again in 120 sec


4:59
Security Verification
Two Factor Authentication
Use fingerprint first or Google Authenticator as backup.
Unlock with Fingerprint
OR USE GOOGLE AUTHENTICATOR
Registered Email
Enter Authenticator Code
369888
Verify Code


5:02
Vault
Search vault...
Edit Secure Source
Source Name
chrome
Website Link
chrome.com
Username
Kunal kudu
Password
Enable AutoFill
Update
Vault
Profile
Settings


5:02
Vault
Search vault...
instagram
Username : kunal kudu
Password : ********
Created : 28 Mar 2026
Updated : 28 Mar 2026
Time : 3 days ago
chrome
Username : Kunal kudu
Password : ********
Created : 29 Mar 2026
Updated : 29 Mar 2026
Time : 2 days ago
Vault
Profile
Settings

Firebase

Hello, Kunal

Welcome back to Firebase!

Get started

Create a new Firebase project
Integrate Firebase products to super-charge your app

Try out a sample app

Build an AI-powered Flutter app
Deploy a sample app that showcases how the Gemini Live API, multimodal prompts, and image creation with Nano Banana all work in Flutter

Try an AI-powered trip planner app
Deploy a sample app that uses Firestore, Authentication, and multimodal input using Firebase AI Logic. Explore the code in Firebase Studio

Try an agentic barista app
Deploy a sample app that uses Firestore, Authentication, and function calling in Firebase AI Logic. Explore the code in Firebase Studio

Search all projects and workspaces

Projects and workspaces

LockBook
lockbook-7bcfc

1 – 1 of 1

Language — English (United States)

Support — Terms — Privacy Policy

Firebase

Search for products

Project Overview

Settings

What's new

Data Connect NEW

Phone Verification NEW

Project shortcuts

Firestore

Authentication

Storage

Product categories

Databases & Storage

Security

AI services

Hosting & Serverless

DevOps & Engagement

Analytics

Spark
No-cost ($0/month)

Upgrade NEW

LockBook

Cloud Firestore ›

Database

(default)

Ask Gemini about the core concepts to use Firestore

Data | Rules | Indexes | Disaster Recovery | Usage | Extensions

Protect your Cloud Firestore resources from abuse, such as billing fraud or phishing Configure App Check

Panel view Query builder

users › 8gFZ9z6eunbs... › vault › KFhb0mKTLPR...

More in Google Cloud

8gFZ9z6eunbs0ObCD3tSb7isruh2

+ Start collection

vault

+ Add field

email: "kudukunal35@gmail.com"

email_otp: "481704"

email_otp_expiry: "2026-03-31T10:53:24.894054"

emergency: "9764605866"

emergency_phone: "9764605866"

name: "kunal kudu "

phone: "9673901910"

profile_image_url: ""

totp_enabled: true

totp_secret: "JBSWY3DPEHPK3PXP"

username: "kunal kudu"

vault

+ Add document

40GgZVq4FCC8xeVzoE7o

82V748pDs8QV2eA0zUs3

KFhb0mKTLPR2R9gLsDxt

N0G3CX4XlAZb3lHa05a3

XazSHEVJKM9vudEF1AU8

KFhb0mKTLPR2R9gLsDxt

+ Start collection

+ Add field

access: true

created: "2026-03-28 10:42:57.915635"

link: "Facebook.com" (string)

pass: "rama1234"

source: "facebook "

updated: "2026-03-29T17:11:15.416"

username: "rama125"

visible: false

Firebase

Search for products

Project Overview

Settings

What's new

Data Connect NEW

Phone Verification NEW

Project shortcuts

Authentication

Firestore

Storage

Product categories

Databases & Storage

Security

AI services

Spark
No-cost ($0/month)

Upgrade NEW

LockBook

Authentication

Users | Sign-in method | Templates | Usage | Settings | Extensions

The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Search by email address, phone number, or user UID

Add user

Identifier	Providers	Created	Signed In	User UID
kiran.gaikwad105@gm...		Mar 31, 2026	Mar 31, 2026	NV3lH6FbSbgB1kRq493Qjjx92...
kunal17@gmail.com		Mar 31, 2026	Mar 31, 2026	jTybBqHxyBbVo1sCnTZoHX1P...
shubhampatil17@gmail...		Mar 31, 2026	Mar 31, 2026	oozib7xuMlcY58CF45g7h4fV9...
onlyforstudies156@gm...		Mar 30, 2026	Mar 30, 2026	oUb83PjUbnZuLgKAFbC9ds3...
parampatil310@gmail....		Mar 30, 2026	Mar 31, 2026	CTH7bWRFJbXk1aHZDTkzjljk...
rama19k@gmail.com		Mar 29, 2026	Mar 29, 2026	cozCQBhwWBQhCvl0mClwo7...
kunal19@gmail.com		Mar 28, 2026	Mar 28, 2026	8fezCQrCgHRc10UybUASLXB...
kuduajay21@gmail.com		Mar 28, 2026	Mar 28, 2026	wL6OfAdf6jVtGO9jQNiJ1jWrb...

Firebase

Search for products

Project Overview

Settings

What's new

Data Connect NEW

Phone Verification NEW

Project shortcuts

Authentication

Firestore

Storage

Product categories

Databases & Storage

Security

AI services

Hosting & Serverless

DevOps & Engagement

Analytics

LockBook

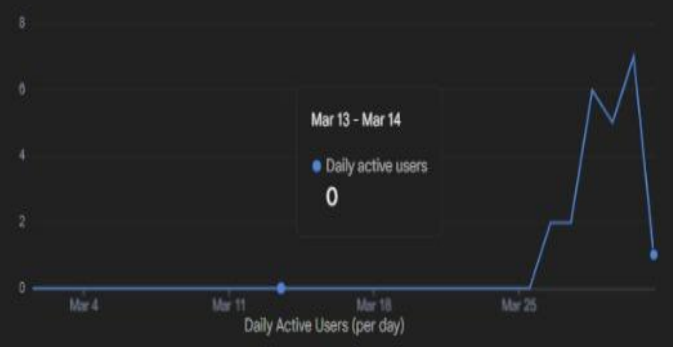
Authentication

Users | Sign-in method | Templates | Usage | Settings | Extensions

Daily Active Users

Current billing period
Feb 28 - Apr 1 (PDT)

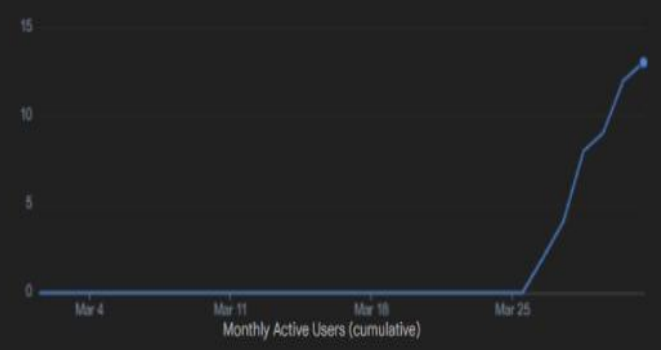
Daily active users
7 peak



Monthly Active Users

Current billing period
Feb 28 - Apr 1 (PDT)

Monthly active users
13



SMS Usage

Current billing period
Feb 28 - Apr 1 (PDT)

Firebase

Search for products

Project Overview

Settings

What's new

Data Connect NEW

Phone Verification NEW

Project shortcuts

Firestore

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(default)

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Panel view Query builder

users › CTH7bWRFJbXk...

More in Google Cloud

(default)

+ Start collection

users

users

+ Add document

8fezCQrCgHRc10UybUASLXBLD743

8gFZ9z6eunbs0ObCD3tSb7isruh2

CTH7bWRFJbXk1aHZDTkzjljkWHE2

NV3lH6FbSbg81kRq493Qjjx92fs1

PzOc7J3anaVkU6sgKMruWDCSXk43

UJ37HdM3KST1VKnkQjcKTKX6BwO2

bH5oqZxETbXLot5r9tF8Pa0xUAj1

cozCQ8hwWBQhCvl0mCIwo7kvWsU2

jTybBqHxyBbVo1sCnTZoHX1P4Or2

oUb83PjUbnZuLgKAFbC9ds3zlSr1

oozib7xuMIcY58CF45g7h4fV9sH2

wL6OfAdf6jVtG09jQNiJ1jWrb8t2

CTH7bWRFJbXk1aHZDTkzjljkWHE2

+ Start collection

vault

+ Add field

```
createdAt: March 30, 2026 at 5:58:11 AM UTC+5:30
email: "parampatil310@gmail.com"
emergency_email: "parampatil310@gmail.com"
emergency_phone: "+917709320548"
phone: "+917709320548"
totp_enabled: false
totp_secret : "JBSWY3DPEHPK3PXP"
username: "param"
```

9. Workflow

Signup / Login
Signup / Login
2FA Verification
2FA Verification
(Fingeprint / Authenticator)
Password Vault
Password Vault
Secure Access
Secure Access Granted
User Dashboard / Vault Access
User Dashboard / Vault Access

Thank You...!!